Current Affairs 30-10-2026

Credit - Drishti Ias

Disinvestment in India

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For Prelims: Disinvestment , Department of Investment and Public Asset Management , Initial Public Offerings , National Monetisation Pipeline , Foreign Direct Investment

For Mains: Disinvestment and privatisation as instruments of economic reform, Role of CPSEs in economic growth and resource mobilisation, Fiscal consolidation and non-debt capital receipts, Transparency and accountability in public asset management

Source: TH

Why in News?

The **Cabinet Committee on Economic Affairs (CCEA)** approved the strategic sale of a 60.72% stake in IDBI Bank to a private entity for Rs 52,878 crore. This landmark deal has positioned the Union Government to comfortably exceed its annual **disinvestment** target for the first time in eight years.

- The government also plans to list other wholly owned entities like the **India Infrastructure Finance Company Ltd (IIFCL),** signalling a renewed, aggressive push toward **disinvestment as a central macroeconomic strategy.**

Summary

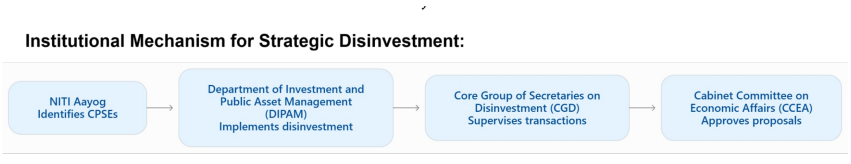
- Disinvestment helps mobilise non-debt capital receipts, improve CPSE efficiency, strengthen corporate governance, and fund infrastructure and social sector investments.
- Transparent, demand-driven reforms, better asset valuation, and international best practices can enhance investor confidence and ensure long-term wealth creation.

What is Disinvestment?

- **About:** Disinvestment refers to the action of an organization or government liquidating, selling, or diluting its equity stake in **Public Sector Enterprises (PSEs),** state-owned financial institutions, or other physical assets.

- It implies the **conversion of public assets into liquid capital,** transitioning ownership either partially or completely into private hands.

- **Objective:** To mobilize **non-debt capital receipts** , reduce the financial burden on the exchequer, and improve corporate governance through market discipline.
- **Nodal Agency:** The **Department of Investment and Public Asset Management (DIPAM)** under the Ministry of Finance is the nodal agency handling all matters related to the sale of Central Government equity.
- **Types of Disinvestment:**
  - **Minority Disinvestment:** The government retains a majority stake (over 51%) and management control, selling a minority share via **Initial Public Offerings (IPOs)** to ensure stock market listing (e.g., **LIC IPO**).
  - **Majority Disinvestment:** The government hands over control to the acquiring entity but retains a minority stake.
  - **Strategic Disinvestment (Privatization):** The transfer of the ownership and management control of a **Central Public Sector Enterprise (CPSE)** to a private strategic buyer (e.g., the 2022 sale of Air India to the Tata Group, and the ongoing IDBI Bank sale).
- **Policy Frameworks:**
  - **New Public Sector Enterprise (PSE) Policy, 2021:** The cornerstone of India's current divestment drive is the **New PSE Policy 2021** , which aims to minimize government presence in business. It classifies the economy into two distinct sectors:
    - **Strategic Sectors:** Include Atomic Energy, Space, Defence; Transport and Telecommunications; Power, Petroleum, Coal; and Banking/Insurance.
      - The government will maintain a **"bare minimum"** presence (one to four companies). The rest will be privatized, merged, or holding-company subsidized.
    - **Non-Strategic Sectors:** Include all other industries (like steel, hospitality). CPSEs in these sectors will either be entirely privatized or shut down.
  - **National Monetisation Pipeline (NMP):** The **National Monetisation Pipeline (NMP)** aims to recycle capital by leasing out mature, brownfield public assets to private players. Following the achievement of 90% of its Phase 1 goals, **NMP 2.0** has set an ambitious target of Rs 16.72 lakh crore for FY2026-2030.
  - **Monetisation of Non-Core Assets:** The **Union Budget 2021–22 proposed the creation of a Special Purpose Vehicle (SPV)** to monetise surplus government land and other non-core assets through sale, concession or similar arrangements.
    - Unlike equity disinvestment, asset monetisation may unlock resources without transferring ownership or management control of an entire public enterprise.
- **Commissions and Recommendations:**
  - **Rangarajan Committee (1993):** Recommended that government ownership in strategic sectors could be diluted up to 49%, and up to 100% in non-strategic sectors.
  - **G.V. Ramakrishna Commission (1996):** Established the core framework for disinvestment, advising a strict classification of CPSEs into strategic, core, and non-core domains.



Why is Disinvestment a Key Economic Strategy?

- **Funding Expenditure:** Disinvestment receipts can unlock capital tied up in underperforming CPSEs and redirect it **towards high-priority social sectors** such as health and education, and infrastructure projects under the NIP.
  - This can support rising government capital expenditure while reducing reliance on additional borrowing and limiting the **risk of crowding out private investment.**
- **Enhancing Operational Efficiency:** State-owned enterprises often face **bureaucratic red tape.**
  - Transferring ownership to the private sector brings in autonomous boards, modern **managerial expertise, and public market accountability, boosting overall profitability.**
- **Democratizing Wealth:** Market listings (IPOs) of CPSEs broaden retail participation in the Indian stock market, allowing citizens to directly own shares in national assets.
  - During the **2014–2019 period, the BSE CPSE Index underperformed significantly with just a 4% average return** , while the benchmark BSE SENSEX surged by 38%. This wide gap in performance served as a primary rationale for **India’s massive structural shift toward aggressive public sector privatization and strategic disinvestment.**
- **Catalyzing 'Privatisation 2.0':** Large-ticket sales shift the focus from mere OFS (which maintains operational status quo) to investor-led, professionally managed enterprises that attract massive **Foreign Direct Investment (FDI).**

Parameter	Liberalization	Disinvestment	Privatization
Core Definition	The relaxation of government regulations, restrictions, and controls on economic activities to allow free-market enterprise.	The sale or liquidation of a portion of the government's equity stake in CPSEs.	The complete transfer of ownership, management, and operational control of a public asset to a private entity.
Primary Objective	To dismantle state monopolies, encourage domestic/foreign investment, and integrate with the global economy.	To raise **non-debt capital receipts** , reduce the fiscal deficit, and unlock trapped asset value without necessarily losing control.	To infuse private capital, introduce corporate efficiency, **eliminate bureaucratic red tape** , and exit non-strategic businesses.
Management & Control	Does not involve ownership transfer. It simply creates a deregulated environment where private players can operate freely.	The government typically retains majority equity (over 51%) and management control (in the case of minority disinvestment).	The government fully cedes management control and majority/total ownership to a private strategic buyer.
Scope of Policy	Macroeconomic policy affecting the entire business ecosystem (tariffs, licenses, FDI limits).	Micro-level financial strategy applied to specific government-owned entities to raise funds or widen public ownership.	Micro-level structural shift applied to specific CPSEs, transferring them completely out of the state apparatus.
Impact on Government Role	State transitions from a "controller" to a "regulator" or "facilitator."	State remains the owner/operator but brings in retail or institutional shareholders for market discipline.	State completely exits the business of running the enterprise.
Indian Examples	Abolition of the 'License Raj' (1991); allowing 100% FDI in the Telecom and Space sectors.	Selling a 3.5% stake in the Life Insurance Corporation (LIC) via an IPO; minor equity dilution in ONGC.	Strategic sale of Air India to the Tata Group (2022); strategic privatization of IDBI Bank (2026).

What are the Concerns Regarding the Disinvestment?

- **The Valuation Conundrum:** Government auditors (**CAG**) often evaluate CPSEs based on their immense underlying land banks and physical assets.
  - However, private bidders value them based on discounted future cash flows (Enterprise Value).
  - This mismatch often leads to stalled bids and political accusations of "cronyism" and underselling.
- **Market Absorption and "Crowding Out":** When the government offloads massive equities (e.g., the mega LIC OFS or offloading SUUTI stakes), it sucks immense liquidity out of the domestic capital markets.
  - This can inadvertently crowd out IPOs of private startups and MSMEs seeking growth capital.
- **Risk of Oligopolistic Rent-Seeking:** Privatizing natural monopolies (such as airports, specific rail routes, or energy grids) without heavily empowering independent regulators (like AERA or **PNGRB**) risks replacing a **public monopoly with a private oligopoly,** leading to predatory consumer pricing.
- **The Legacy Land Trap:** Disentangling non-core assets, particularly leasehold land allotted by State Governments to CPSEs, leads to protracted legal disputes (e.g., the decades-long VSNL land demerger issue) that derail transaction timelines.

What Measures are Needed to Reform India’s Disinvestment Policy to Improve Efficiency and Asset Value?

- **Transition to a Demand-Based Approach:** The government should prioritize PSE sales based on early investor feedback and strong bids at expected valuations, abandoning rigid, government-led selection criteria.
- **Rolling Three-Year Privatisation Stream:** Establishing a multi-year pipeline based on strategic priorities allows prospective investors ample time for **due diligence** , resulting in deeper market participation and **better price discovery** .
- **Calibrated Disinvestment as an Interim Measure:** Because full privatization is a lengthy, complex journey, the government should adopt **calibrated disinvestment** .
  - Bringing the government’s stake down to 51% across 95 listed PSEs could yield approximately **Rs 9 lakh crore** , while maintaining the state as the single largest shareholder.
- **Ring-Fencing Proceeds:** A statutory framework must ensure that disinvestment receipts are strictly channelled into capital asset creation (via bodies like the **National Investment and Infrastructure Fund**) rather than plugging everyday revenue deficits.
- **International Best Practices:** India's framework can be optimized by adopting proven international sovereign asset management models:
  - **The Temasek Model (Singapore):** Singapore manages its state-owned enterprises through Temasek Holdings, an independent investment company. Temasek operates strictly on commercial principles to maximize shareholder value, **completely separating the state’s regulatory role from its ownership role.**
  - **SASAC Model (China):** The State-owned Assets Supervision and Administration Commission consolidates state assets under one umbrella, aggressively pushing mergers and acquisitions to create globally competitive mega-corporations before partial public listing.

Conclusion

Strategic disinvestment should shift from short-term revenue generation to efficient asset management and wealth creation. Transparent, demand-driven privatisation can strengthen fiscal prudence, improve CPSE efficiency, and fund essential social and infrastructure investments.

Drishti Mains Practice Questions

How can India ensure that disinvestment proceeds contribute to long-term capital formation rather than short-term fiscal management?

Frequently Asked Questions (FAQs)

1. What is disinvestment, and what is its primary objective?
Disinvestment involves the sale or dilution of government equity in public sector enterprises. Its objectives include mobilising non-debt capital receipts, improving efficiency, and reducing the financial burden on the government.

2. Which institution is responsible for managing the Union Government's disinvestment process?
The Department of Investment and Public Asset Management (DIPAM) under the Ministry of Finance is the nodal department for managing the Union Government's equity sales in CPSEs.

3. What is the difference between disinvestment and strategic disinvestment?
Disinvestment may involve selling a minority stake without transferring control. Strategic disinvestment involves transferring ownership and management control to a strategic buyer.

4. What is the objective of the New Public Sector Enterprise Policy, 2021?
The policy seeks to minimise the government's presence in commercial activities, retain a bare minimum presence in strategic sectors, and consider privatisation or closure in non-strategic sectors.

[Watch Video on YouTube: ▶ **https://www.youtube.com/embed/0YnCiahAeCE**]

UPSC Civil Services Examination, Previous Year Question (PYQ)

Q1. In the context of governance, consider the following: (2010)

1. Encouraging Foreign Direct Investment inflows
2. Privatization of higher educational Institutions
3. Down-sizing of bureaucracy
4. Selling/offloading the shares of Public Sector Undertakings

Which of the above can be used as measures to control the fiscal deficit in India?

(a) 1, 2 and 3
(b) 2, 3 and 4
(c) 1, 2 and 4
(d) 3 and 4 only

Ans: (d)

Q2. Why is the Government of India disinvesting its equity in the Central Public Sector Enterprises (CPSEs)? (2011)

1. The Government intends to use the revenue earned from the disinvestment mainly to pay back the external debt.
2. The Government no longer intends to retain the management control of the CPSEs.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Ans: (d)

Q3. With reference to the National Investment Fund to which the disinvestment proceeds are routed, consider the following statements: (2010)

1. The assets in the National Investment Fund are managed by the Union Ministry of Finance.
2. The National Investment Fund is to be maintained within the Consolidated Fund of India.
3. Certain Asset Management Companies are appointed as the fund managers.
4. A certain proportion of annual income is used for financing select social sectors.

Which of the statements given above is/are correct?

(a) 1 and 2
(b) 2 only
(c) 3 and 4
(d) 3 only

Ans: (c)

Mineral Exchange

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Source: TH

Why in News?

The Ministry of Mines notified the **Mineral Exchange Rules, 2026** to formalize the trading of physical minerals. Following this, the **Indian Bureau of Mines (IBM)** invited applications to establish and operate these exchanges. In response, PTC India Ltd. (formerly Power Trading Corporation of India Ltd.) board approved an equity investment to incorporate a **wholly-owned subsidiary to set up a Mineral Exchange.**

What is the Mineral Exchange?

- **About:** It is a centralized, electronic marketplace designed to facilitate the transparent and competitive trading of physical mineral ores.
  - Unlike existing financial commodity stock exchanges that trade derivative contracts, Mineral exchange focuses on **delivery-based spot contracts where buyers and sellers trade actual physical mineral quantities.**
- **Mechanism:** It marks a paradigm shift in the mining sector, transitioning from a traditional "one-to-many" sales model (such as state-run auctions) to a competitive "many-to-many" trading platform.
- **Legal Bedrock:** Section 18B of the **Mines and Minerals (Development and Regulation) (MMDR) Act, 1957** empowers the Central Government to promote mineral markets and facilitate trading of minerals, concentrates and processed products, including metals, through mineral exchanges.
- **Supervising Body:** The **Indian Bureau of Mines (IBM), a subordinate office under the Ministry of Mines,** has been designated as the **regulatory authority** responsible for registering and overseeing mineral exchanges.
  - This regulatory mandate covers all minerals except those listed under **Part A (coal and lignite) and Part B (atomic minerals)** of the First Schedule of the Mines and Minerals (Development and Regulation) Act, 1957.
  - Commodity derivatives, on the other hand, remain regulated by the **Securities and Exchange Board of India (SEBI).**
- **Initial Commodities:** The exchange will initially focus on **physical ores rather than processed metals** , beginning with bulk commodities such as iron ore, bauxite, limestone, and manganese ore. Iron ore is expected to be the primary driver due to its massive scale and liquidity.
- **Future Scope:** Subsequent phases are expected to include **critical minerals like lithium, cobalt, nickel, and rare earth** elements to support India's electric vehicle, renewable energy, and defense sectors.
- **Global Precedents:**
  - **London Metal Exchange (LME):** Serves as the primary global benchmark for trading base and industrial metals.
  - **Dalian Commodity Exchange (DCE):** Based in China, it operates as a major global exchange

specializing in iron ore.

Need for a Dedicated Mineral Exchange in India

- **Transparent Price Discovery:** The current system lacks a national benchmark price for many minerals (e.g., iron ore, bauxite). Prices often vary drastically between states like Odisha, Karnataka, and Chhattisgarh.
  - An exchange would create a transparent, demand-supply-driven **National Mineral Index** , preventing monopolistic pricing by large mining cartels.
- **Boosting GDP Contribution:** The mining sector's contribution to India's GDP is stagnant at around **1.5% - 2%** .
  - The **National Mineral Policy 2019** **targets increasing this significantly.** An exchange ensures better **realization of value, augmenting state exchequers through fair royalty calculations.**
- **Strategic Management of Critical Minerals:** With the push for EVs and the **National Green Hydrogen Mission** , minerals like lithium, cobalt, and **rare earth elements (REEs)** are crucial.
  - An exchange can help domestic industries secure these minerals at competitive, transparent prices, reducing import dependency.
- **Credit Access and Financialization:** Standardized warehouse receipts from a Mineral Exchange can be used as **collateral by MSMEs and downstream industries to access formal credit, bridging a massive working capital gap.**
- **Curbing Illegal Mining:** By mandating strict origin-tracing and assaying (quality testing) for traded minerals, an exchange can **indirectly choke the black market and illegal mining operations,** ensuring only legally extracted ores enter the organized supply chain.
- **Empowering Smaller Miners:** Small and medium miners currently suffer from limited bargaining power compared to large corporate buyers. An exchange provides a level playing field.
- **Establishing an Asian Benchmark:** As the global race for mineral security intensifies, India's Mineral Exchanges could evolve beyond a local platform to provide a common reference point for mineral prices across Asia - a region that accounted for roughly **USD 1.37 trillion in critical-mineral imports and USD 757 billion in exports in 2024.**

What are the Key Bottlenecks in Establishing a Mineral Exchange in India?

- **Infrastructure Deficit:** Physical delivery requires a massive network of accredited warehouses, railway sidings, and bulk handling ports. Currently, India's logistics cost (around 13-14% of GDP) makes physical mineral trading highly localized.
- **Generating Liquidity:** The success of any exchange depends on high trading volumes. The platforms must attract **enough buyers and sellers to make participation commercially viable.**
- **Preventing Market Manipulation** : Regulators must actively monitor the market to prevent a small number of large producers or buyers from dominating the trade, coordinating bids, or weakening competition.
- **Grade Standardization and Assaying: Minerals are heterogeneous.** For example, iron ore grades vary from 50% Fe to 65% Fe.
  - Standardizing contracts for an exchange requires an independent, world-class assaying and testing infrastructure, which is currently lacking.
- **Reluctance of Major Players:** State-owned enterprises (like NMDC) and large private steelmakers who

currently benefit from long-term captive contracts may resist transitioning to a **spot-exchange model that disrupts their established supply chains.**

- **Avoiding Compliance Overlap:** The new platforms must complement existing mining regulations rather than burdening participants with redundant compliance requirements.

Conclusion

The successful implementation of mineral exchanges requires a phased rollout, unified regulatory framework, blockchain-based traceability, upgraded assaying infrastructure and suitable amendments to the MMDR Act to ensure market liquidity and global competitiveness.

Drishti Mains Question:

How can mineral exchanges contribute to India's critical mineral security and strategic autonomy amid growing global competition for mineral resources?

Frequently Asked Questions (FAQs)

1. What are the Mineral Exchange Rules, 2026?
They provide a regulatory framework for establishing and operating mineral exchanges to facilitate transparent, competitive and delivery-based trading of physical minerals.

2. Which provision of the MMDR Act, 1957 provides for mineral exchanges?
Section 18B empowers the Central Government to promote mineral markets and trading through mineral exchanges.

3. Which authority regulates mineral exchanges in India?
The Indian Bureau of Mines (IBM), under the Ministry of Mines, is responsible for registering and overseeing mineral exchanges. Commodity derivatives remain under SEBI's jurisdiction.

4. Which minerals are expected to be traded initially through mineral exchanges?
The initial focus will be on bulk minerals such as iron ore, bauxite, limestone and manganese ore, through spot and delivery-based contracts.

5. How can mineral exchanges strengthen India's critical mineral security?
They can facilitate transparent trading of lithium, cobalt, nickel and rare earth elements, improve domestic market access and support EVs, renewable energy and defence manufacturing.

[Watch Video on YouTube: ▶ https://www.youtube.com/embed/KCVqcAxXoJU]

UPSC Civil Services Examination, Previous Year Questions (PYQs)

Prelims

Q. With reference to the management of minor minerals in India, consider the following statements: (2019)

1. Sand is a ‘minor mineral’ according to the prevailing law in the country
2. State Governments have the power to grant mining leases of minor minerals, but the powers regarding the formation of rules related to the grant of minor minerals lie with the Central Government.
3. State Governments have the power to frame rules to prevent illegal mining of minor minerals.

Which of the statements given above is/are correct?

(a) 1 and 3 only

(b) 2 and 3 only

(c) 3 only

(d) 1, 2 and 3

Ans: (a)

Q. What is/are the purpose/purposes of ‘District Mineral Foundations’ in India? (2016)

1. Promoting mineral exploration activities in mineral-rich districts
2. Protecting the interests of the persons affected by mining operations
3. Authorizing State Governments to issue licences for mineral exploration

Select the correct answer using the code given below:

(a) 1 and 2 only

(b) 2 only

(c) 1 and 3 only

(d) 1, 2 and 3

Ans: (b)

Mains

Q. Despite India being one of the countries of Gondwanaland, its mining industry contributes much less to its Gross Domestic Product (GDP) in percentage. Discuss. **(2021)**

Operation Trident

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Source: PIB

The **Narcotics Control Bureau (NCB)** , along with the **Mizoram Police and Enforcement Directorate (ED)** , launched **Operation Trident** against a suspected **drug trafficking network** across **Assam, Mizoram, and Manipur.**

- **About Operation Trident: Operation Trident** aims to break entire drug networks by targeting their **money and supply chains** , instead of focusing only on drug seizures.
  - It is part of the **Vision Document on Narcotics Control 2026-2029** , which aims to eliminate drug networks by **31st December 2029** .
  - The operation reflects a strategic shift from simple drug seizures to **crippling the financial and corporate-style infrastructure** of cartels.
    - The cartel allegedly moved **methamphetamine and heroin** across India, Myanmar and Bangladesh through the **“Death Triangle Corridor.”**
- **About Narcotics Control Bureau :** NCB is India’s top coordinating **agency for drug law enforcement** .
  - It was constituted by the Government of India in **1986** under the **Narcotic Drugs and Psychotropic Substances Act, 1985.**
    - The **National Policy on Narcotic Drugs and Psychotropic Substances** is based on **Article 47 of the Constitution** , which directs the State to prohibit intoxicating drugs harmful to health except for medicinal use.
  - It operates under the **Ministry of Home Affairs.**
  - India is a signatory to key international conventions like **Single Convention on Narcotic Drugs, 1961** (as amended by 1972 Protocol), **Convention on Psychotropic Substances, 1971** , and **UN Convention against Illicit Traffic in Narcotic Drugs and Psychotropic Substances, 1988.**

About Enforcement Directorate (ED):

- The ED is a **multi-disciplinary organization** mandated with investigation of offences of **money laundering** and **violations of foreign exchange laws.**
- It functions under the **Department of Revenue of the Ministry of Finance.**
- Its headquarter is in **New Delhi.**

Read More: Narcotics Control Bureau and Operation Global-Hunt

Karela Lake Uncovers India's Monsoon History

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Source: DD India

A study reconstructed **4,600 years** of climate and environmental history in the **Central Ganga Plain,** showing how the **Indian Summer Monsoon influenced lakes, vegetation and early farming** .

- **Study Site:** Sediments from **Karela Lake** , an **oxbow lake near Hulas Khera in Lucknow district, Uttar Pradesh** were used.
  - The study identified 4 phases of climate event i.e. **4.2-thousand-year arid event, Late Holocene, Roman Warm Period and Little Ice Age** .
- **Conducted By:** The **Birbal Sahni Institute of Palaeosciences (BSIP)** , an autonomous institute under the **Department of Science and Technology (DST)** .
  - BSIP was founded in **1946** to promote research in palaeobotany and received **UNESCO support (1951–53)** and became an **autonomous body in 1969** .
- **Methods Used:** Scientists used **radiocarbon dating** **, carbon and nitrogen isotopes, grain-size analysis, sediment modelling and pollen analysis** .
- **Key Findings:** Monsoon changes influenced **lake water levels, sediment transport, vegetation, microbial activity and agricultural practices** .

Oxbow Lake

- An oxbow lake is a **U-shaped or crescent-shaped water body formed when a river meander becomes separated from its main channel** .
  - It typically develops in **flat, low-lying plains where rivers form winding loops** . Erosion along the outer bank and deposition of sediments along the inner bank gradually widen the meander, narrowing the strip of land between its two ends, known as the neck.
- During floods or periods of high flow, the river may cut through this narrow neck, creating a shorter channel.
  - Deposition subsequently seals off the **abandoned loop, forming an oxbow lake** . Over time, the **lake accumulates sediments, organic matter and vegetation, gradually evolving into a swamp or marsh.**
  - Eventually, it **may dry up completely, leaving behind a crescent-shaped depression known as a meander scar.**
- Notable Examples in India are **Kanwar Lake (Bihar)** , **Gogabeel Lake (Bihar)** , **Surah Tal (UP).**



Read More: **Encroachment at Karbatal Wetland**

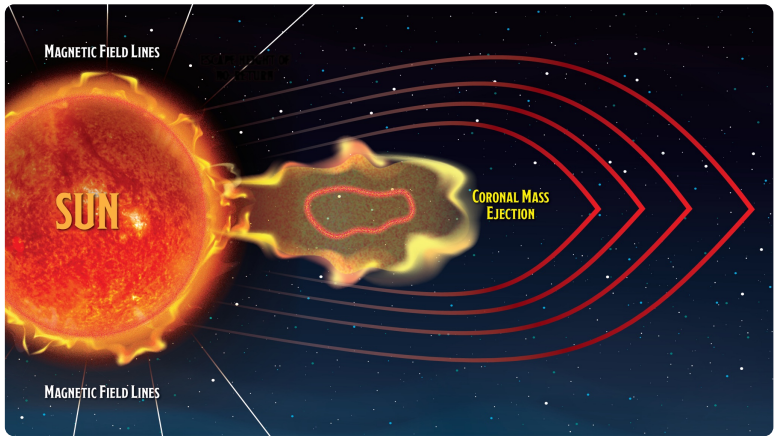
Aditya-L1 and Proba-3 Advance Coronal Mass Ejection Studies

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Source: TH

Scientists used **observations from India's Aditya-L1 and the European Space Agency's (ESA) Proba-3 to jointly study Coronal Mass Ejections (CMEs)** close to the Sun.

- **Instruments Used:** Aditya-L1's **Visible Emission Line Coronagraph (VELC)** provided spectroscopic data, while Proba-3's **ASPIICS coronagraph** provided detailed images of the solar corona.
  - Combining both instruments helped estimate the **mass, density, shape, dynamics and energy of CMEs** more accurately than using a single spacecraft.
- **Key Finding:** The two eruptions showed different patterns of **expansion and energy distribution** , helping scientists understand the early evolution of CMEs.
- **About Missions:**
  - **Aditya-L1** : It is the **first space-based observatory-class Indian solar mission** to study the **Sun** .
    - The spacecraft is planned to be placed in a **halo orbit** around the **Lagrangian point1 (L1)** of the Sun-Earth system.
    - A satellite placed in L1 point can **continuously view the Sun without any occultation/eclipse.**
  - **Proba-3** : This mission belongs to the **European Space Agency (ESA)** which is the **world's first precision formation-flying mission** .
    - It was launched by **ISRO's PSLV-C59 rocket** in 2024.
    - It uses **two satellites flying 150 meters apart** to create an artificial solar eclipse and study the **Sun's faint corona.**
- **Coronal Mass Ejections (CMEs):** They are **large eruptions of solar plasma** and **magnetic fields into space** , often associated with solar flares.
  - They occur when **unstable magnetic structures on the Sun release trapped material** and are most frequent during **solar maximum** .
  - If directed towards Earth, CMEs can **disrupt satellites, threaten astronauts and disturb Earth's magnetic field** through magnetic reconnection.
  - This process directs charged particles towards the poles, where they interact with atmospheric gases, producing **colourful auroras.**



Read More: **Aditya-L1 Observations of Coronal Mass Ejection - ISRO**